Data Governance, (Compliance) and Ethics



WEEK 02: DATA GOVERNANCE AND MANAGEMENT (PART 2)

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Overview of course content

1. Data Governance and Management (Weeks 1-3)
2. Privacy and Data Protection (Weeks 4-5)
3. Ethical Issues Pertaining to Data (Weeks 6, 8-9)*
4. Fairness, Accountability, and Transparency of Algorithmic Systems (Weeks 10-12)

* Week 7 = Reading Week (No classes)

Agenda

Week 1: Introduction to Data Governance (DG)
- Definitions of DG
- Data Governance & Data Management
- Good and Poor DG
- Introduction to DMBOK wheel

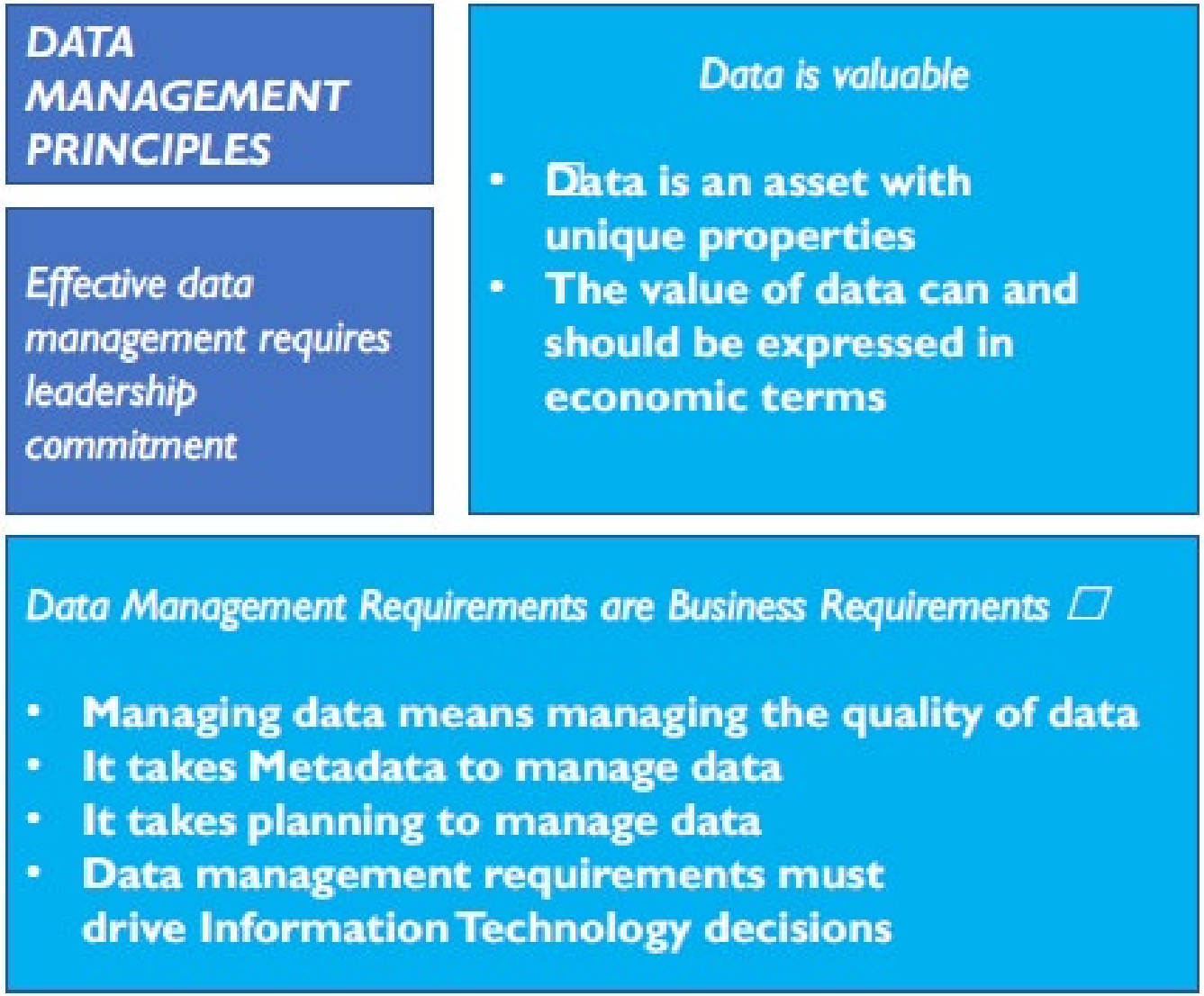
Week 2: Application of concept
- Management of data strategies, policies, processes and procedures
- Data Lifecycle and Data Types
- Data Quality, Data Provenance, Data Integrity
- Supplementary Material: Data Management principles

Week 3: DG Implementation
- Roles and responsibilities
- DG operating model
- DG maturity levels
- Ladley's 8-phase implementation process for DG

Supplementary Material
- Data risk identification and management
- Models and tools for data governance

Data Management Principles 1/2

- DG shares characteristics with other forms of asset management
- Strategic and operational needs must be balanced.
- Striking a balance requires to follow Data Management principles, which may include:



DMBOK Figure 1 Data Management Principles

Data Management Principles 1/2

DATA MANAGEMENT PRINCIPLES

Effective data management requires leadership commitment

Data is valuable

- Data is an asset with unique properties
- The value of data can and should be expressed in economic terms

Data Management Requirements are Business Requirements

- Managing data means managing the quality of data
- It takes Metadata to manage data
- It takes planning to manage data
- Data management requirements must drive Information Technology decisions

DMBOK Figure 1 Data Management Principles



Properties:

- Data is not consumed when it is used
- Data is not tangible; but it is durable
- Data does not wear out through its value changes as it ages
- Data is easy to copy and transport
- Data can be stolen without being gone; but it is not easy to reproduce if lost or destroyed
- Data is dynamic and can be used for multiple purposes and multiple people at the same time

Data Management Principles 1/2

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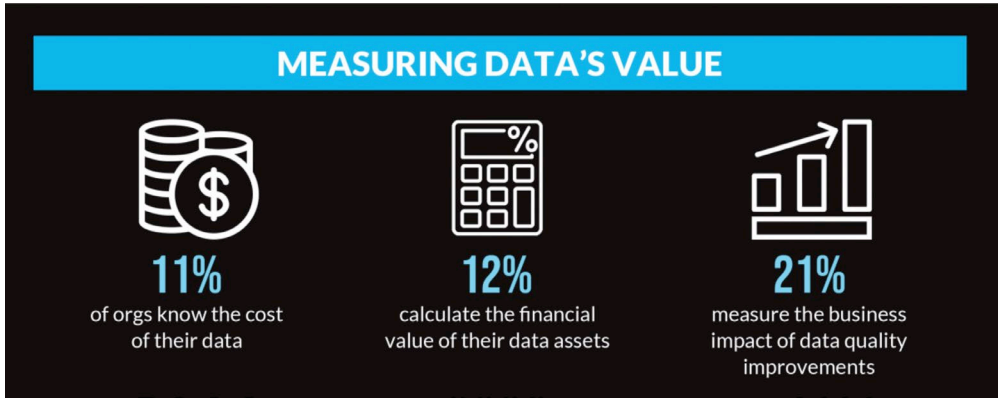
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DMBOK Figure 1 Data Management Principles

- There are no standards only techniques based on measures of information value
  - Foundational: info mngt practices
  - Financial: financial impact of info in your business



Data Management Principles 1/2

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DMBOK Figure 1 Data Management Principles



Source: Kaggle, Delta Partners analysis



Data Management Principles 1/2

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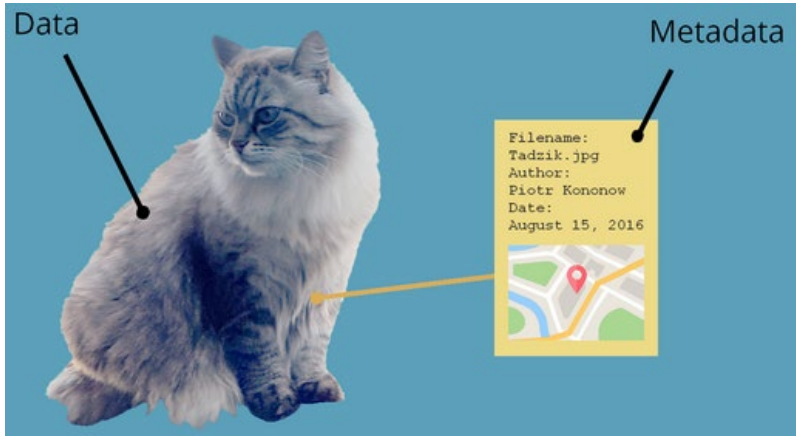
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DMBOK Figure 1 Data Management Principles





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DMBOK Figure 1 Data Management Principles



- Planning from an architectural and process perspective is needed to coordinate work and keep the end results aligned

Data Management Principles 1/2

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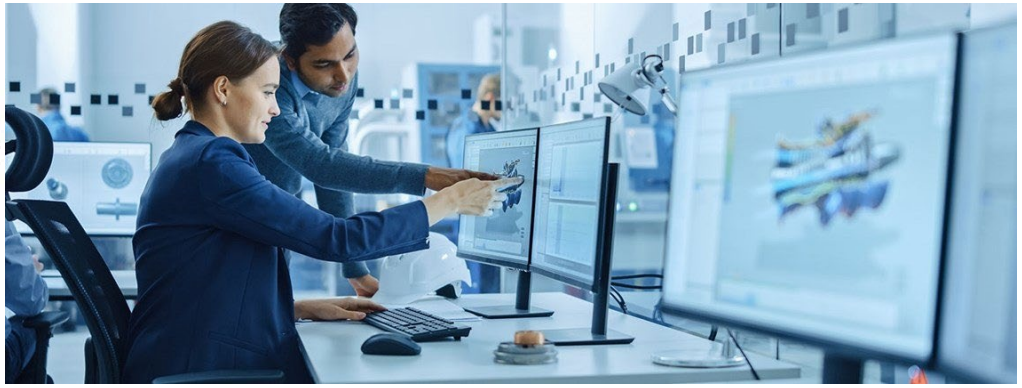
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DMBOK Figure 1 Data Management Principles



- Ensure that technology serves, rather than drives, an organization's strategic data needs

Data Management Principles 2/2

Data Management depends on diverse skills

- Data management is cross-functional
- Data management requires an enterprise perspective
- Data management must account for a range of perspectives

Data Management is lifecycle management

- Different types of data have different lifecycle characteristics
- Managing data includes managing the risks associated with data



- DM requires a range of skills and expertise (technical and non-technical)
- Data is fluid. DM must constantly evolve to keep up with the ways data is created and used and the data consumers who use it

DMBOK Figure 1 Data Management Principles

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DMBOK Figure 1 Data Management Principles

Data Life cycle



- DM practices have to recognize the differences in management requirements and be flexible enough to meet different kinds of data lifecycle requirements.
- Data-related risks must be managed as part of the data lifecycle